

Homework 1

Due: 22nd January, 2026

Exercise 1.1 (Hogg, Tanis and Zimmerman Exercise 2.4.7). Suppose that 2000 points are selected independently and at random from the unit square $\{(x, y) : 0 \leq x < 1, 0 \leq y < 1\}$. Let W equal the number of points that fall into $A = \{(x, y) : x^2 + y^2 < 1\}$.

1. How is W distributed?
2. Give the mean, variance and standard deviation of W .
3. What is the expected value of $\frac{W}{500}$?
4. Use the computer to select 2000 pairs of random numbers. (This is pretty easy on excel - something ChatGPT is really good at haha) Determine the value of W and use that value to find an estimate for π . (Of course, we know the real value of π and more will be said about estimation later in this text.)
5. How could you extend part (d) to estimate the volume $V = \frac{4}{3}\pi$ of a ball of radius 1 in 3-space?
6. How could you extend these techniques to estimate the volume of a ball of radius 1 in n -space?

Sol. The area of the quarter circle is $\frac{\pi}{4}$ and that of the unit box is 1; the success probability of a single point X_i landing in A is given by

$$\mathbb{P}(X_i \in A) = \frac{\pi}{4}$$

so recording 1 for a point landing in A and 0 otherwise, $X_i \sim \text{Ber}\left(\frac{\pi}{4}\right)$. Now, $W = \sum_{i=1}^{2000} X_i$ and each X_i is an independent $\text{Ber}\left(\frac{\pi}{4}\right)$ random variable, so $W \sim \text{Bin}\left(2000, \frac{\pi}{4}\right)$.

Using the formulae from class,

$$\begin{aligned} \mathbb{E}[W] &= 2000 \cdot \frac{\pi}{4} = 500\pi, & \text{Var}(W) &= 2000 \cdot \frac{\pi}{4} \cdot \left(1 - \frac{\pi}{4}\right) = 500\pi \left(1 - \frac{\pi}{4}\right), \\ \sigma(W) &= \sqrt{\text{Var}(W)} = \sqrt{500\pi \left(1 - \frac{\pi}{4}\right)}. \end{aligned}$$

Hence,

$$\mathbb{E}\left[\frac{W}{500}\right] = \frac{1}{500}\mathbb{E}[W] = \frac{500\pi}{500} = \pi.$$

W is an estimator for π ! I ran a couple of different iterations (see attached excel notebook), and got two samples for W : 3.118 and 3.214.

A similar method could estimate the volume of a ball of radius 1 in 3-space. Taking the part of the ball $\{(x, y, z) : x^2 + y^2 + z^2 < 1\}$ that lives in $[0, 1]^3$ as A , we would find that the success probability of a single point landing in A is

$$\mathbb{P}(X_i \in A) = \frac{4\pi}{3} \cdot \frac{1}{8} = \frac{\pi}{6},$$

since there are eight quadrants in $\mathbb{R}^3$ and hence A is an eighth of the full ball. Thus, the total number of points in A is distributed according to $\text{Bin}\left(2000, \frac{\pi}{6}\right)$, with $\mathbb{E}[W] = \frac{2000\pi}{6}$. We then would see that $\frac{W}{250}$ is a good estimate for $\frac{4\pi}{3}$, the volume of the ball in 3-space.

More generally, taking the part of the ball $\{\vec{x} : \sum_{i=1}^n x_i^2 < 1\}$ that lives in $[0, 1]^n$ as A , we would find that the success probability of a single point landing in A is

$$\mathbb{P}(X_i \in A) = \frac{\text{vol}(\text{ball})}{2^n}$$

since there are 2^n quadrants in $\mathbb{R}^n$. Thus, the total number of points in A is distributed according to $\text{Bin}\left(2000, \frac{\text{vol}(\text{ball})}{2^n}\right)$, with $\mathbb{E}[W] = \frac{2000}{2^n} \text{vol}(\text{ball})$. We then would see that $\frac{2^n W}{2000}$ is a good estimate for the volume of the ball in n -space.

Exercise 1.2 (Panaretos, Exercises 4-5). 1. Let $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$ be independent Poisson random variables with parameters $\lambda, \mu > 0$ respectively. Show that $X + Y \sim \text{Poisson}(\lambda + \mu)$.

2. Conclude that if $X_1, X_2, \dots, X_n$ are i.i.d. $\text{Poisson}(\lambda)$ random variables, then

$$\sum_{i=1}^n X_i \sim \text{Poisson}(n\lambda).$$

3. Let X and Y be as in the first part of this exercise. Show that the conditional distribution of X given $X + Y = k$ is $\text{Binom}\left(k, \frac{\lambda}{\lambda + \mu}\right)$, i.e.

$$\mathbb{P}(X = m | X + Y = k) = \binom{k}{m} \left(\frac{\lambda}{\lambda + \mu}\right)^m \left(1 - \frac{\lambda}{\lambda + \mu}\right)^{k-m}$$

for all $m \in \{0, 1, 2, \dots, k\}$.

Proof. For the first part, notice via independence that

$$\mathbb{P}(X + Y = m) = \sum_{k=0}^m \mathbb{P}(X = k \text{ and } Y = m - k) = \sum_{k=0}^m \mathbb{P}(X = k) \mathbb{P}(Y = m - k).$$

Using the probability mass function of a Poisson random variable, we find

$$\mathbb{P}(X + Y = m) = \sum_{k=0}^m e^{-\lambda} \frac{\lambda^k}{k!} e^{-\mu} \frac{\mu^{m-k}}{(m-k)!} = \frac{e^{-(\lambda+\mu)} m!}{\sum_{k=0}^m \frac{m!}{k!(m-k)!} \lambda^k \mu^{m-k}}$$

Via the binomial theorem, we find

$$\mathbb{P}(X + Y = m) = \frac{e^{-(\lambda+\mu)}}{m!} \sum_{k=0}^m \binom{m}{k} \lambda^k \mu^{m-k} = \frac{e^{-(\lambda+\mu)}}{m!} (\lambda + \mu)^m$$

which is precisely the probability mass function of a $\text{Poisson}(\lambda + \mu)$ random variable. Hence, $X + Y \sim \text{Poisson}(\lambda + \mu)$.

This immediately implies the second part; applying this iteratively, we find that

$$\sum_{i=1}^n X_i \sim \text{Poisson}(\lambda + \lambda + \dots + \lambda) = \text{Poisson}(n\lambda).$$

We combine these results with the definition of conditional expectation to obtain the third part. We observe that

$$\mathbb{P}(X = m | X + Y = k) = \frac{\mathbb{P}(X = m \text{ and } X + Y = k)}{\mathbb{P}(X + Y = k)} = \frac{\mathbb{P}(X = m \text{ and } Y = k - m)}{\mathbb{P}(X + Y = k)} = \frac{\mathbb{P}(X = m) \mathbb{P}(Y = k - m)}{\mathbb{P}(X + Y = k)}$$

where we have used that $X = m$ and $X + Y = k$ implies $Y = k - m$. Since $X \sim \text{Poisson}(\lambda)$, $Y \sim \text{Poisson}(\mu)$ and $X + Y \sim \text{Poisson}(\lambda + \mu)$ we have

$$\begin{aligned} \mathbb{P}(X = m | X + Y = k) &= \frac{e^{-\lambda} \frac{\lambda^m}{m!} e^{-\mu} \frac{\mu^{(k-m)}}{(k-m)!}}{e^{-(\lambda+\mu)} \frac{(\lambda+\mu)^k}{k!}} = \frac{k!}{m!(k-m)!} \frac{\lambda^m \mu^{k-m}}{(\lambda+\mu)^k} \\ &= \binom{k}{m} \frac{\lambda^m}{(\lambda+\mu)^m} \frac{\mu^{k-m}}{(\lambda+\mu)^{k-m}} = \binom{k}{m} \left(\frac{\lambda}{\lambda+\mu}\right)^m \left(\frac{\mu}{\lambda+\mu}\right)^{k-m}, \end{aligned}$$

which is the desired result using $1 - \frac{\lambda}{\lambda+\mu} = \frac{\mu}{\lambda+\mu}$. $\square$

Exercise 1.3 (Anderson, Seppäläinen and Valkó Exercises 2.67-8). 1. Show that if $X \sim \text{Geom}(p)$, then

$$\mathbb{P}(X = n + k | X \geq n) = \mathbb{P}(X = k), \quad n, k \geq 0.$$

This could be called the *memoryless property* of the geometric distribution, because it states the following. Given that there are no successes in the first n trials, the probability that there are exactly k more failures before the first success (i.e. at exactly $n + k$ failures before the first success) is the same as the probability that a freshly started sequence of trials yields the first success right after trial k . In other words, the earlier n failures are forgotten.

Note: the specific version I wrote down is slightly different from the ASV text, because our version of a geometric random variable is defined in a slightly different way (allowing the value zero) than it is in the ASV text.

2. Suppose that X is a random variable whose possible values are $\{0, 1, 2, 3, \dots\}$ and satisfies $0 < \mathbb{P}(X = 0) := p < 1$ and the memoryless property above. Show that X must be a geometric random variable for some parameter p .

Hint: use $\mathbb{P}(X = k) = \mathbb{P}(X = k + 1 | X \geq 1)$ repeatedly.

Proof. For the first part, we compute directly with the definition of conditional probability that

$$\mathbb{P}(X = n + k | X \geq n) = \frac{\mathbb{P}(X = n + k \text{ and } X \geq n)}{\mathbb{P}(X \geq n)} = \frac{\mathbb{P}(X = n + k)}{\mathbb{P}(X \geq n)}$$

where we have simplified the numerator by dropping the redundant condition $X \geq n$ (since $X = n + k \implies X \geq n$). Now, using the probability mass function, we find

$$\frac{\mathbb{P}(X = n + k)}{\mathbb{P}(X \geq n)} = \frac{p(1-p)^{n+k}}{\sum_{m=n}^{\infty} \mathbb{P}(X = m)} = \frac{p(1-p)^{n+k}}{\sum_{m=n}^{\infty} p(1-p)^m} = \frac{p(1-p)^{n+k}}{\frac{p(1-p)^n}{1-(1-p)}} = p(1-p)^k = \mathbb{P}(X = k),$$

as desired.

For the second part, we first define $p := \mathbb{P}(X = 0)$. We will use induction and the memoryless property to show that $X \sim \text{Geom}(p)$.

For our base case, $\mathbb{P}(X = 0)$ agrees with the probability mass function of a $\text{Geom}(p)$ random variable for $k = 0$ by definition. For the inductive step, assume that it is true for some $k \geq 0$, i.e. $\mathbb{P}(X = k) = p(1-p)^k$. Then, using the memoryless property with $n = 1$ we find

$$p(1-p)^k = \mathbb{P}(X = k) = \mathbb{P}(X = k + 1 | X \geq 1) = \frac{\mathbb{P}(X = k + 1 \text{ and } X \geq 1)}{\mathbb{P}(X \geq 1)} = \frac{\mathbb{P}(X = k + 1)}{\mathbb{P}(X \neq 0)}.$$

We have simplified the numerator by dropping the redundant assumption $X \geq 1$, since $X = k + 1 \implies X \geq 1$. We have simplified the denominator by using that $X \geq 1 \iff X \neq 0$, since the sample space is $\{0, 1, 2, 3, \dots\}$. So,

$$p(1-p)^k = \frac{\mathbb{P}(X = k + 1)}{1 - \mathbb{P}(X = 0)} = \frac{\mathbb{P}(X = k + 1)}{1 - p} \implies \mathbb{P}(X = k + 1) = p(1-p)^{k+1}.$$

Hence, by induction, $\mathbb{P}(X = k) = p(1-p)^k$ for all $k \geq 0$. This is precisely the probability mass function of a $\text{Geom}(p)$ random variable, so $X \sim \text{Geom}(p)$, as desired. $\square$